

Internal note: is the LSR boundary-driven threshold 0.347 (NeurIPS) actually the correct analog of the LSE saddle-driven threshold 0.6226?

Drafted as a reference for referee responses to the AAAI submission.

Anticipated referee question

The AAAI submission compares two thermodynamic-limit capacity numbers under the exact spherical density:

- LSE (this work): $\alpha_c^{\text{LSE}} \approx 0.6226$, derived from a saddle of the spurious-sum integrand at the golden ratio $\varphi^* = (\sqrt{5} - 1)/2$.
- LSR (NeurIPS companion [1]): $\alpha_{\text{th}}^{\text{LSR}} \approx 0.347$, derived from the support-entry condition $M \cdot \Pr[\varphi_{1\mu} > \varphi_c] \simeq 1$.

A careful referee may object that these are not the same kind of quantity. The NeurIPS number is the threshold at which the *first* spurious pattern enters the LSR kernel support, computed from a single-closest / extreme-value argument. The AAAI number is the threshold at which the *cumulative* spurious sum saturates the retrieval term, computed from a saddle of the integrand. If for LSR the cumulative contribution from the band of patterns near φ_c overtakes the single closest pattern, one might worry that $\alpha_{\text{th}}^{\text{LSR}} = 0.347$ is the wrong number to compare against $\alpha_c^{\text{LSE}} = 0.6226$, and that the LSR cumulative-band saturation threshold is different.

This note shows that, in the thermodynamic limit $N \rightarrow \infty$, the two LSR criteria *coincide*: NeurIPS's 0.347 is simultaneously the support-entry threshold and the boundary-driven cumulative-saturation threshold, and it is therefore the correct apples-to-apples analog of the LSE 0.6226. The finite- N correction is $O(\ln N/N)$ and explicitly computed below.

1 The two parallel derivations

The spurious sum at the retrieval state $\mathbf{x} = \boldsymbol{\xi}^1$ is

$$S = \sum_{\mu \neq 1} K(\varphi_{1\mu}) \approx M \int_{-1}^1 f(\varphi) K(\varphi) d\varphi, \quad f(\varphi) = C_N (1 - \varphi^2)^{(N-3)/2}, \quad (1)$$

where K is the kernel and $C_N = 1/B((N-1)/2, 1/2)$. We compare with the retrieval term $K_{\text{ret}} = K(1)$ and call the breakdown the condition $S \simeq K_{\text{ret}}$.

LSE. Kernel $K_{\text{LSE}}(\varphi) = \exp(\beta_{\text{net}} N \varphi)$, retrieval value $K_{\text{ret}} = e^{\beta_{\text{net}} N}$. The integrand exponent (per unit N) is

$$g_{\text{LSE}}(\varphi) = \frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}} \varphi. \quad (2)$$

Stationarity at $\beta_{\text{net}} = 1$ gives $\varphi^{*2} + \varphi^* - 1 = 0$, so $\varphi^* = (\sqrt{5} - 1)/2$ and (using $1 - \varphi^{*2} = \varphi^*$)

$$g_{\text{max}} = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774. \quad (3)$$

Setting $\alpha N + g_{\text{max}} N = \beta_{\text{net}} N$ yields the saddle-driven LSE threshold

$$\boxed{\alpha_c^{\text{LSE}} = \beta_{\text{net}} - g_{\text{max}} \approx 0.6226.} \quad (4)$$

LSR. Kernel $K_{\text{LSR}}(\varphi) = \max(0, 1 - b + b\varphi) = b(\varphi - \varphi_c) \mathbf{1}_{\varphi \geq \varphi_c}$ with $\varphi_c = (b - 1)/b$. Retrieval value $K_{\text{ret}} = K_{\text{LSR}}(1) = 1$. The integrand on $[\varphi_c, 1]$ is $f(\varphi) b(\varphi - \varphi_c)$. Taking logarithms, the exponent (per unit N) is

$$h_{\text{LSR}}(\varphi) = \underbrace{\frac{1}{2} \ln(1 - \varphi^2)}_{\text{entropy, } O(1)} + \underbrace{\frac{1}{N} \ln[b(\varphi - \varphi_c)]}_{O(1/N), \text{subleading}}. \quad (5)$$

To leading order in N only the entropy contributes. The entropy $\frac{1}{2} \ln(1 - \varphi^2)$ is monotonically decreasing on $[\varphi_c, 1]$ (its global maximum at $\varphi = 0$ lies *outside* the LSR support), so the integrand-exponent maximum is at the left boundary $\varphi = \varphi_c$ with value

$$h_{\text{max}} = \frac{1}{2} \ln(1 - \varphi_c^2). \quad (6)$$

Setting $\alpha N + h_{\text{max}} N = 0$ yields the boundary-driven LSR threshold

$$\boxed{\alpha_{\text{th}}^{\text{LSR}} = -h_{\text{max}} = -\frac{1}{2} \ln(1 - \varphi_c^2)}. \quad (7)$$

For $b = 2 + \sqrt{2}$ this is $\alpha_{\text{th}}^{\text{LSR}} = \frac{1}{2} \ln 2 \approx 0.347$, which is exactly the NeurIPS number.

2 Why the numbers (0.6226 vs. 0.347) look so different

Both formulas are instances of the same identity:

$$\boxed{\alpha_c = (\text{retrieval rate}) - (\text{integrand maximum value})}. \quad (8)$$

The contrast in the numerical answers has two ingredients, not one.

1. **Where the integrand maximum sits.** LSE's exponential kernel adds the linear term $\beta_{\text{net}}\varphi$ to the entropy, pushing the maximum inward to an interior saddle at the golden ratio $\varphi^* = 0.618$. LSR's compactly-supported kernel contributes only an $O(1/N)$ correction; the leading-order maximum sits at the support boundary $\varphi_c = 0.707$ for $b = 2 + \sqrt{2}$.
2. **Retrieval-kernel scale.** LSE's $K_{\text{ret}} = e^{\beta_{\text{net}} N}$ supplies a rate β_{net} on the right-hand side of the balance. LSR's $K_{\text{ret}} = 1$ supplies a rate of 0. Algebraically: $\alpha_c^{\text{LSE}} = \beta_{\text{net}} - g_{\text{max}}$ and $\alpha_{\text{th}}^{\text{LSR}} = 0 - h_{\text{max}}$.

The values inserted into the boxed identity are:

	retrieval rate	integrand max position	integrand max value	threshold
LSE (this work)	$\beta_{\text{net}} = 1$	$\varphi^* = (\sqrt{5} - 1)/2$	$g_{\text{max}} = +0.3774$	0.6226
LSR (NeurIPS)	0	$\varphi_c = (b - 1)/b$	$h_{\text{max}} = -0.347$	0.347

The 0.6226-vs-0.347 contrast is therefore neither a saddle-vs-boundary distinction alone nor a kernel-scale distinction alone: it is both.

3 Cumulative band vs. single closest, the original worry

The referee's intuition is correct that there is a band of patterns just above φ_c whose cumulative contribution exceeds that of the single closest pattern. The point of this section is that the threshold extracted from the cumulative-band calculation coincides with the threshold extracted from the single-closest / support-entry calculation in the thermodynamic limit.

Position of the cumulative band. The full LSR integrand is $f(\varphi)(\varphi - \varphi_c)$. Setting its log-derivative to zero,

$$-\frac{(N-3)\varphi}{1-\varphi^2} + \frac{1}{\varphi - \varphi_c} = 0, \quad (9)$$

and expanding around φ_c with $\delta = \varphi - \varphi_c \ll 1$ gives the interior maximum at

$$\delta_\star \approx \frac{1 - \varphi_c^2}{N \varphi_c} = O(1/N). \quad (10)$$

So LSR *does* have a formal interior maximum of the integrand, but its position is glued to the support boundary in the $N \rightarrow \infty$ limit. The Gaussian width of the saddle is $\sigma \sim 1/\sqrt{N}$.

Cumulative-band sum. Laplace integration around $\delta_\star$ gives

$$S_{\text{LSR}} \sim M b f(\varphi_c) \cdot \frac{1}{N^2 \varphi_c^2} = \exp[(\alpha - \alpha_{\text{th}}^{\text{LSR}}) N] \cdot \frac{b\sqrt{2}}{N^{3/2} \sqrt{\pi} \varphi_c} \quad (11)$$

(using $f(\varphi_c) \sim \sqrt{N/(2\pi)} e^{-\alpha_{\text{th}}^{\text{LSR}} N}$ at leading order). Setting $S_{\text{LSR}} = K_{\text{ret}} = 1$:

$$\alpha_c^{\text{cum, LSR}}(N) = \alpha_{\text{th}}^{\text{LSR}} + \frac{\frac{3}{2} \ln N + \frac{1}{2} \ln \frac{\pi}{2} + \ln(\varphi_c/b)}{N} = \alpha_{\text{th}}^{\text{LSR}} + O\left(\frac{\ln N}{N}\right). \quad (12)$$

For $N = 25$ and $b = 2 + \sqrt{2}$ this gives a shift of about $+0.15$ relative to the leading-order value; for $N = 100$ it is about $+0.05$; in the thermodynamic limit it vanishes.

Single-closest argument. NeurIPS works with the typical maximum overlap among M patterns: $\varphi_{\text{max}}(\alpha) \approx \sqrt{1 - e^{-2\alpha}}$ (from inverting $M \int_{\varphi_{\text{max}}}^1 f d\varphi = 1$). The condition $\varphi_{\text{max}} = \varphi_c$ gives precisely the leading-order value of (12), with no $O(\ln N/N)$ correction. The single-closest argument is the leading-order term of the cumulative one.

Why they agree at $N \rightarrow \infty$. Because the LSR integrand maximum sits *at* the support boundary (unlike the LSE case, where the saddle is in the interior at $\varphi^* = 0.618$), the band of patterns and the single closest pattern are not statistically separable structures: the band collapses onto the boundary as $N \rightarrow \infty$, the width $\delta_\star \rightarrow 0$ as $1/N$, and the number of patterns inside the band stays $O(1)$ at the threshold. For LSE the saddle width is also $\sigma \sim 1/\sqrt{N}$ but its *position* is bounded away from any boundary (at $\varphi^* \approx 0.618$), so the cumulative-band picture survives the thermodynamic limit and is qualitatively distinct from any single-closest argument.

4 Bottom line for the referee response

There is no separate ‘‘cumulative-band breakdown threshold for LSR’’ distinct from NeurIPS’s $\alpha_{\text{th}}^{\text{LSR}} = 0.347$ in the thermodynamic limit. $\alpha_{\text{th}}^{\text{LSR}} = 0.347$ is simultaneously

1. the support-entry threshold (first pattern enters the LSR kernel support), and
2. the boundary-driven cumulative-saturation threshold (spurious sum first balances the retrieval term),

to leading order in N . Its finite- N correction is the $+O(\ln N/N)$ shift in (12), which appears as a small additive offset in the apparent boundary at the moderate N accessible to honest Monte Carlo and vanishes in the $N \rightarrow \infty$ limit. Comparing 0.6226 (LSE, this work) and 0.347 (LSR, NeurIPS) is therefore an apples-to-apples comparison of thermodynamic-limit, exact-density, first-noise-saturation thresholds. The numerical contrast reflects both the interior-saddle vs. boundary-max distinction (the kernel-shape effect) and the $e^{\beta_{\text{net}} N}$ vs. 1 scaling of the retrieval term (the kernel-amplitude effect).

References

- [1] T. Petrova, E. Polyachenko, R. State. *Retrieval Metastability via Kramers Escape and Centroid Hopping*. NeurIPS, 2026 (placeholder bib key).